

Table 5: Transfer learning ability of LPW. Comparison of performance with different pre-train dataset ImageNet and LPW.

Matching rate(%)	PRID 2011				iLIDS-VID			
Rank	1	5	10	20	1	5	10	20
RQEN+ImageNet	91.8	98.4	99.3	99.8	77.1	93.2	97.7	99.4
RQEN+LPW	93.4	98.4	99.2	100.0	80.0	94.4	98.2	99.3

Table 6: Cross-dataset testing performance on PRID 2011. CD (?) is trained on Shinhuhkan 2014 dataset and other methods are trained on iLIDS-VID. The PRID2011 is used for testing.

Matching rate(%)	PRID 2011			
Rank	1	5	10	20
RQEN	61.8	82.6	90.4	96.1
QAN	34.0	61.3	74.0	83.1
Batch normalization	34.3	61.2	76.4	88.8
CNN+RNN	28	57	69	81
CD (?)	17	-	43	52

Table 7: Comparison with state-of-the-art methods on MARS dataset.

Matching rate(%)	MARS			
Rank	1	5	20	mAP
our+XQDA+Rerank	77.83	88.84	94.29	71.14
our	73.74	84.9	91.62	51.70
TriNet+Reranking	83.03	93.69	97.63	66.43
IDE+Reranking	73.94	-	-	68.45
CNN+SRM+TAM	70.6	90.0	97.6	50.7
CNN+XQDA	65.0	81.1	88.9	49.3

LPW dataset is of great help to initial training in pedestrian re-identification.

Complementary Information Extraction

We generated the quality scores of the test set using RQEN used for intuitive exploration. Some partially occluded images are selected for evaluation. Figure 7 provides a vivid illustration of how the scores reflect the corresponding regional quality. The quality of images' regions with some noise such as occlusion and blur is poor and the corresponding scores are lower.

Figure 1 clearly demonstrates the advantages of RQEN in the aggregation of sequence features. Compared with the average pooling method, it can make more effective use of complementary information in different images' parts of the sequence. The effective information in the region of one frame can make up for missing portions of the same region's information in other frames. The more robust feature representation can be extracted via RQEN with the assist of partial quality scores and get better performance.

Conclusion

This paper proposes the Region-based Quality Estimation Network (RQEN) for video-based person re-identification. RQEN can learn partial quality of each image and aggregate



Figure 7: The quality scores of selected images' different parts. The scores in [0,1] are mapped to the colors in the transition from blue to red.

the images' complementary partial information of different frames in an image sequence. We design an end-to-end training strategy with ingenious gradient design and jointly train the network with classification and verification signal. The RQEN leads to the state-of-the-art results on PRID 2011 and iLIDS-VID. We also propose a large-scale and clean dataset named "Labeled Pedestrian in the Wild (LPW)" which contains 7,694 tracklets with over 590,000 images. The dataset, which has occlusion, large posture change and huge age span, is more challenging and suitable for practical use.

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